

1. Analyze the convergence of the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{3n+2}$

Solution. This series has some negative terms, so we first consider the same series but with all positive terms. Since

$$\lim_{n \rightarrow \infty} \frac{\frac{1}{3n+2}}{\frac{1}{n}} = \lim_{n \rightarrow \infty} \frac{n}{3n+2} = \frac{1}{3},$$

the Limit Comparison Test shows that

$$\sum_{n=1}^{\infty} \frac{1}{3n+2} \quad \text{and} \quad \sum_{n=1}^{\infty} \frac{1}{n}$$

have the same behavior, and by the p -test, $\sum_{n=1}^{\infty} 1/n$ diverges. Thus the series we're given does not converge absolutely.

Since $1/(3n+2)$ decreases to zero and $(-1)^n/(3n+2)$ is alternating, we can conclude by the alternating series test that our series does converge, and thus is conditionally convergent.

2. Analyze the convergence of the series $\sum_{n=1}^{\infty} \frac{n}{3^n}$.

Solution. We see the exponential term and use the ratio test:

$$\lim_{n \rightarrow \infty} \left| \frac{\frac{n+1}{3^{n+1}}}{\frac{n}{3^n}} \right| = \lim_{n \rightarrow \infty} \left| \frac{n+1}{n} \cdot \frac{3^n}{3^{n+1}} \right| = \lim_{n \rightarrow \infty} \frac{n+1}{n} \cdot \frac{1}{3} = \frac{1}{3}.$$

Since this is less than 1, we conclude that the series converges absolutely by the ratio test.

3. Find the radius and interval of convergence of the power series $\sum_{n=1}^{\infty} \frac{2^n}{n} x^n$. Use that

$$\lim_{n \rightarrow \infty} \left| \frac{\frac{2^{n+1}}{n+1} x^{n+1}}{\frac{2^n}{n} x^n} \right| = 2|x|.$$

Solution. The above is the limit computation in the ratio test, and so the series will converge absolutely provided $2|x| < 1$, or in other words provided that $|x| < 1/2$. Thus the radius of convergence is $1/2$ and we have absolute convergence on the interval $(-1/2, 1/2)$. To completely determine the interval of convergence, we must check the behavior at the endpoints. At $x = -1/2$, the series is

$$\sum_{n=1}^{\infty} \frac{2^n}{n} \cdot \left(\frac{-1}{2} \right)^n = \sum_{n=1}^{\infty} \frac{(-1)^n}{n},$$

which converges by the alternating series test. At $x = 1/2$, the series is

$$\sum_{n=1}^{\infty} \frac{2^n}{n} \cdot \left(\frac{1}{2} \right)^n = \sum_{n=1}^{\infty} \frac{1}{n},$$

which diverges by the p -test. Thus the interval of convergence is $[-1/2, 1/2)$.

4. Using a series you already know, find the Taylor series for $f(x) = xe^{4x}$.

Solution. We know that $e^x = \sum_{n=0}^{\infty} \frac{1}{n!} x^n$, so plugging in $4x$ and multiplying by x gives

$$xe^{4x} = x \sum_{n=0}^{\infty} \frac{1}{n!} (4x)^n = x \sum_{n=0}^{\infty} \frac{4^n}{n!} x^n = \sum_{n=0}^{\infty} \frac{4^n}{n!} x^{n+1}.$$